

Weighting Scheme: Hierarchical Risk Parity (HRP)

MetaFVG Mean-Reversion Portfolio – 14-Instrument Diversified Universe

August 1, 2026

1 Scheme Description

The portfolio blends 14 instruments trading a Spearman-gated MetaFVG mean-reversion strategy, deliberately assembled across different central banks and currency blocs (AUDUSD, EURUSD, GBPUSD, NZDUSD, USDCAD, USDCHF, USDJPY, US500, GER40, JP225, HK50, USDSGD, EURNOK, USDZAR). The current production blend uses equal weights ($w_i = 1/14$ for all i). This report tests *Hierarchical Risk Parity* (HRP, López de Prado 2016, “Building Diversified Portfolios that Outperform Out-of-Sample”), which replaces the quadratic-optimization step of mean-variance and risk-parity methods with a tree-structured allocation derived from the correlation matrix’s hierarchical clustering. HRP was designed to be robust to estimation error in the covariance matrix (no matrix inversion) by allocating weight through a dendrogram rather than solving a global optimization.

1.1 Algorithm

1. Distance matrix. Convert the correlation matrix ρ into a proper metric distance,

$$d_{ij} = \sqrt{\frac{1}{2}(1 - \rho_{ij})}.$$

2. Hierarchical clustering. Condense d into its upper-triangular form and apply single-linkage agglomerative clustering to build a dendrogram over the 14 instruments.

3. Quasi-diagonalization. Read the leaf order off the dendrogram and reorder the instruments by it, so that instruments merged early in the tree (more correlated) sit adjacent to one another in the resulting sequence.

4. Recursive bisection. Walk down the tree top-down. At each split of a cluster c into c_0 (first half of the quasi-diagonal order) and c_1 (second half), each sub-cluster’s variance is computed using *inverse-variance allocation within the sub-cluster* (diagonal of the covariance matrix only, HRP’s standard simplification – no sub-cluster covariance terms):

$$\text{IVP}_i = \frac{1/\Sigma_{ii}}{\sum_{k \in c} 1/\Sigma_{kk}}, \quad V(c) = \text{IVP}^\top \Sigma_c \text{IVP}.$$

The split factor allocates *more* weight to the lower-variance side:

$$\alpha = 1 - \frac{V(c_0)}{V(c_0) + V(c_1)}, \quad w_{c_0} *= \alpha, \quad w_{c_1} *= (1 - \alpha).$$

This repeats recursively until every cluster is a singleton, giving the final weight vector w (renormalized to sum to 1).

Implementation note: all covariance/variance products above were computed via elementwise (Hadamard) broadcast-and-sum, e.g. `(np.outer(ivp, ivp) * sub).sum()`, rather than matrix multiplication, and the covariance matrix itself was built from a hand-rolled pairwise-complete estimator rather than `DataFrame.cov()` – both `DataFrame.cov()` and `Series.cov()` were found to crash this environment’s BLAS/LAPACK backend (exit 127, no traceback), in addition to the previously known `np.corrcoef` / `matrix-matmul` crash mode. `DataFrame.corr()` itself is unaffected and was used directly for ρ .

2 Quasi-Diagonalized Cluster Order

The dendrogram leaf order groups instruments by realized correlation structure:

```
EURNOK -- JP225 -- US500 -- GER40 -- GBPUSD -- EURUSD -- AUDUSD -- NZDUSD --
          USDJPY -- HK50 -- USDCAD -- USDZAR -- USDCHF -- USDSGD
```

Several groupings match fundamental intuition: the three equity indices **JP225**, **US500**, **GER40** land adjacent despite trading different central-bank regimes and sessions – consistent with a shared global risk-on/risk-off factor dominating their correlation. **AUDUSD** and **NZDUSD**, the classic commodity-bloc pair, are adjacent as expected. **GBPUSD** and **EURUSD**, the two large European majors, are adjacent as well. **EURNOK** anchors one end of the order (Norway’s oil-linked currency, least like the rest of the book) and **USDSGD** anchors the other. **HK50** sits with **USDJPY** rather than with the other equity indices – a less intuitive pairing, likely an artifact of single-linkage clustering “chaining” through a weak intermediate correlation rather than a strong direct relationship.

3 HRP Weights

Table 1: HRP weights, sorted descending. Ratio column is $w_i/(1/14)$.

Symbol	Weight	Weight / (1/14)
USDSGD	0.206 04	2.885
USDCAD	0.144 79	2.027
EURUSD	0.124 27	1.740
EURNOK	0.096 92	1.357
GBPUSD	0.086 97	1.218
USDCHF	0.078 41	1.098
USDJPY	0.072 36	1.013
AUDUSD	0.048 18	0.675
NZDUSD	0.044 77	0.627
USDZAR	0.028 34	0.397
US500	0.022 46	0.314
GER40	0.022 23	0.311
HK50	0.013 93	0.195
JP225	0.010 33	0.145
Sum	1.000 00	14.000

HRP’s recursive bisection pushes weight heavily toward the lower-variance FX legs – USDSGD, USDCAD, EURUSD, and EURNOK all land at $1.4\text{--}2.9\times$ equal weight – and away from the four equity indices, which are squeezed down to roughly $0.15\text{--}0.31\times$ equal weight (JP225 and HK50 the most, at $\sim 0.15\text{--}0.20\times$). This mirrors the direction seen with ERC (equity indices underweighted), but is considerably more extreme here because HRP’s top-level bisection groups three of the four equity indices together early in the quasi-diagonal order, and inverse-variance allocation then compounds against that whole cluster relative to the large low-variance FX block on the other side of the split.

4 Correlation Impact

Table 2: Realized correlation and diversification impact, equal-weight vs. HRP.

Metric	Equal Weight	HRP
Weighted avg. pairwise correlation	0.015 615	0.017 022
Diversification ratio (DR)	2.080 922	1.538 298

HRP does not reduce realized correlation here – the weighted average pairwise correlation *rises* from 0.0156 to 0.0170 (a $\sim 9\%$ relative increase), because the scheme concentrates weight in a cluster of FX pairs (USD-quoted majors/crosses) that are, on net, slightly *more* mutually correlated than the full 14-instrument book, while pulling weight away from the equity indices that anchor much of the book’s diversification. The diversification ratio also falls sharply ($2.081 \rightarrow 1.538$): DR rewards spreading weight across instruments with high standalone volatility relative to blended portfolio volatility, and HRP’s variance-minimizing objective does the opposite by construction.

5 Results

6 Takeaway

HRP fails on every axis this exercise cares about. The weighted average pairwise correlation *increases* ($0.0156 \rightarrow 0.0170$) instead of falling, the diversification ratio drops by nearly a quarter ($2.081 \rightarrow 1.538$), and Sharpe collapses by two-thirds ($0.518 \rightarrow 0.170$) alongside similarly large drops in Sortino, Calmar, and recovery factor. The mechanism is clear from the weights table: HRP’s variance-based recursive bisection has no notion of expected return and systematically defunds the equity-index legs – JP225 down to $0.15\times$ and HK50 to $0.20\times$ equal weight – purely because their standalone P&L variance is higher, even though those same instruments are exactly the ones providing this book’s fundamental (cross-asset-class) diversification. On this data, equal weighting remains clearly superior; HRP should not replace it for this portfolio.

Table 3: Quantstats metrics, equal-weight vs. HRP blend.

Metric	Equal Weight	HRP
Sharpe	0.517 941	0.169 721
Sortino	0.816 977	0.264 785
Calmar	0.343 017	0.106 463
CAGR	0.000 183	0.000 057
Max Drawdown (%)	−5.325 228	−5.325 228
Volatility (%)	3.527 622	3.343 598
Tail Ratio	1.238 689	1.002 700
Ulcer Index	0.000 192	0.000 295
Kelly Criterion	0.066 763	0.027 229
VaR (%)	−0.358 268	−0.344 199
CVaR (%)	−0.715 011	−0.797 777
Skew	1.544 839	1.976 561
Recovery Factor	2.260 123	0.701 973
Win rate	0.308 850	0.298 013
Best day	0.000 291	0.000 291
Worst day	−0.000 239	−0.000 239
Total return	0.001 204	0.000 374